

AI for Bharat Hackathon

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Team Name : ScriptDevs

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Problem Statement : AI for Rural Innovation & Sustainable Systems

Brief about the Idea:

Bridging the Digital Divide with Voice-First Intelligence

RUGRO transforms a simple phone into a powerful personal assistant. By replacing complex menus with natural conversation, we eliminate barriers of literacy and bandwidth for rural communities.

Voice-to-UI Workflow



Users speak in their native tongue (Hindi, regional dialects). Rugro doesn't just reply, it navigates them directly to specific transaction screens.

Low-Bandwidth Optimization



Engineered specifically for 2G/3G connectivity, ensuring accessibility in remote areas where data-heavy apps fail.

Adaptive Intelligence



Centralized AI with Long-Term Memory remembers history and preferences to provide personalized "nudges" for savings and health.

Trusted & Secure



Moves rural users from "searching for answers" to "completing tasks" through a trusted, secure, and human-centric interface.

Goal: To move the rural user from "searching for answers" to "completing tasks" through a trusted, secure, and human-centric interface

Prototype Scope Across Rural Workflows

Agriculture



crop support, contextual
ask, attachment-guided
farming help

Market



listings, buyer matches,
pricing context, contact
workflows

Economics



loan eligibility, savings
guidance, insurance
claims

Health

crop support, contextual
ask, attachment-guided
farming help

Knowledge

learning resources and
community guidance

Trust + Profile

saved user details, memory,
auth, and reuse across flows

Why AI is required in your solution?

In **RUGRO**, AI is not a feature, it is the **primary user interface**. For rural users, traditional menus are a barrier; AI replaces them with a natural, voice-first experience that mimics human conversation.

01

Multilingual

Fluidity allows the system to detect and switch between Hindi, English, and regional dialects instantly for natural interaction.

02

Intent & Entity

Extraction distinguishes if a user is asking about crop prices, health symptoms, or subsidy eligibility.

03

Memory-Backed

Context leverages user history to provide personalized, continuous support instead of isolated interactions.

04

Friction Removal replaces rigid menus with a 'Zero-UI' approach, making digital services accessible to users regardless of literacy levels.

Without AI, Rugro collapses into a maze of static forms and rigid menus. AI is the only technology capable of removing the friction that currently prevents rural communities from adopting digital services. It moves the user from "navigating an app" to "having a conversation that gets results."

How AWS services are used within your architecture?

Cloud Infrastructure

(AWS Compute)

- ECS Fargate
- Application Load Balancer
- CloudWatch Monitoring

AI Intelligence Layer

(Amazon Bedrock)

Nova Micro – Speed Layer
Claude – Reasoning Layer
Intent routing & context

Voice Pipeline

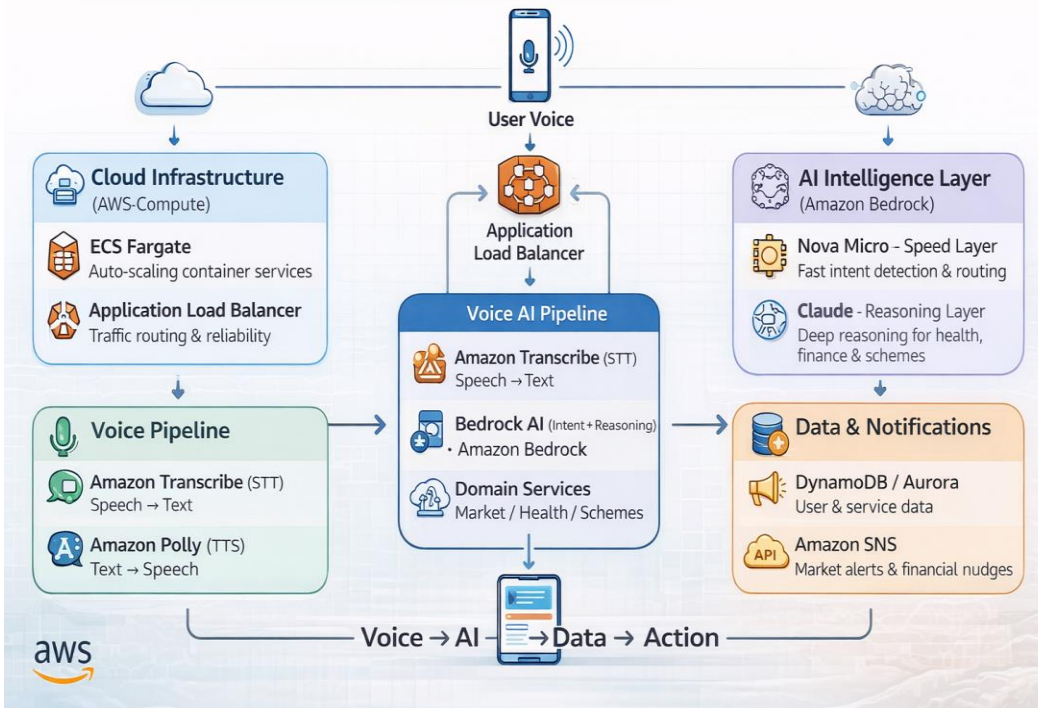
- Amazon Transcribe (STT)
- Amazon Polly (TTS)
- Amazon Translate

Data & Notifications

DynamoDB / Aurora
Market & scheme data
Amazon SNS alerts

AWS Architecture Backbone

Scalable cloud infrastructure powering Rugro's voice-first rural assistant



What value the AI layer adds to the user experience?

Zero-UI Friction:

Collapses a complex multi-service app into a single, natural conversation, removing the literacy and navigation barriers of traditional menus.

Grounded Context Awareness:

The AI "sees" what the user sees by analyzing real-time screen data and photo attachments (like crop pests or medical reports) to provide relevant, non-generic advice.

Cognitive Load Reduction:

Simplifies decision-making by moving users directly from a spoken question to a concrete outcome, ensuring faster task completion even on low-bandwidth networks.

Speech-to-Action Workflows:

Moves beyond text replies by programmatically triggering UI changes, automatically opening market charts, health flows, or training videos based on voice intent.

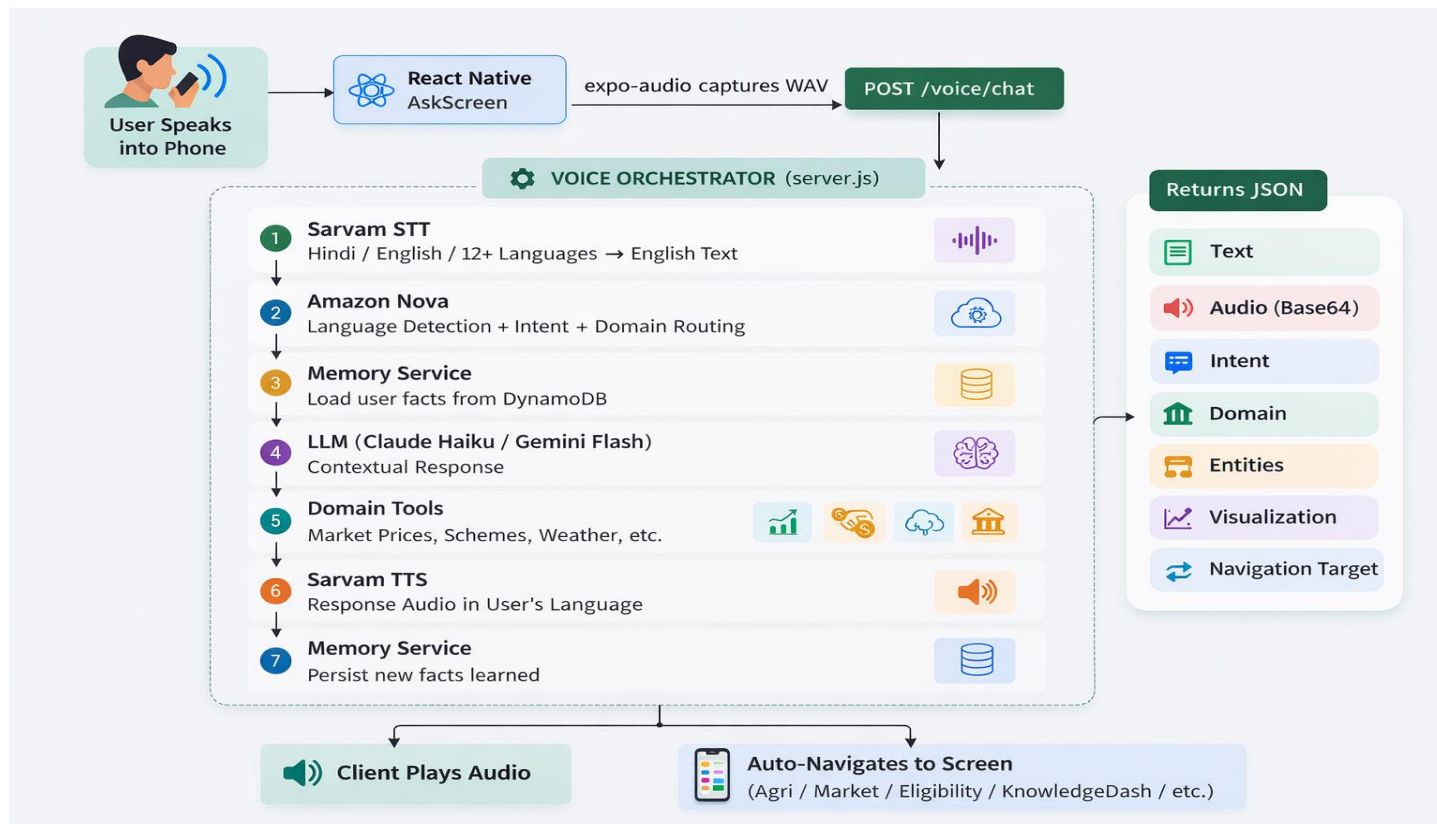
Proactive Personal Assistant:

Leverages persistent memory to remember user history (crop cycles, health records, and preferences), eliminating the need for repetitive form-filling or data entry.

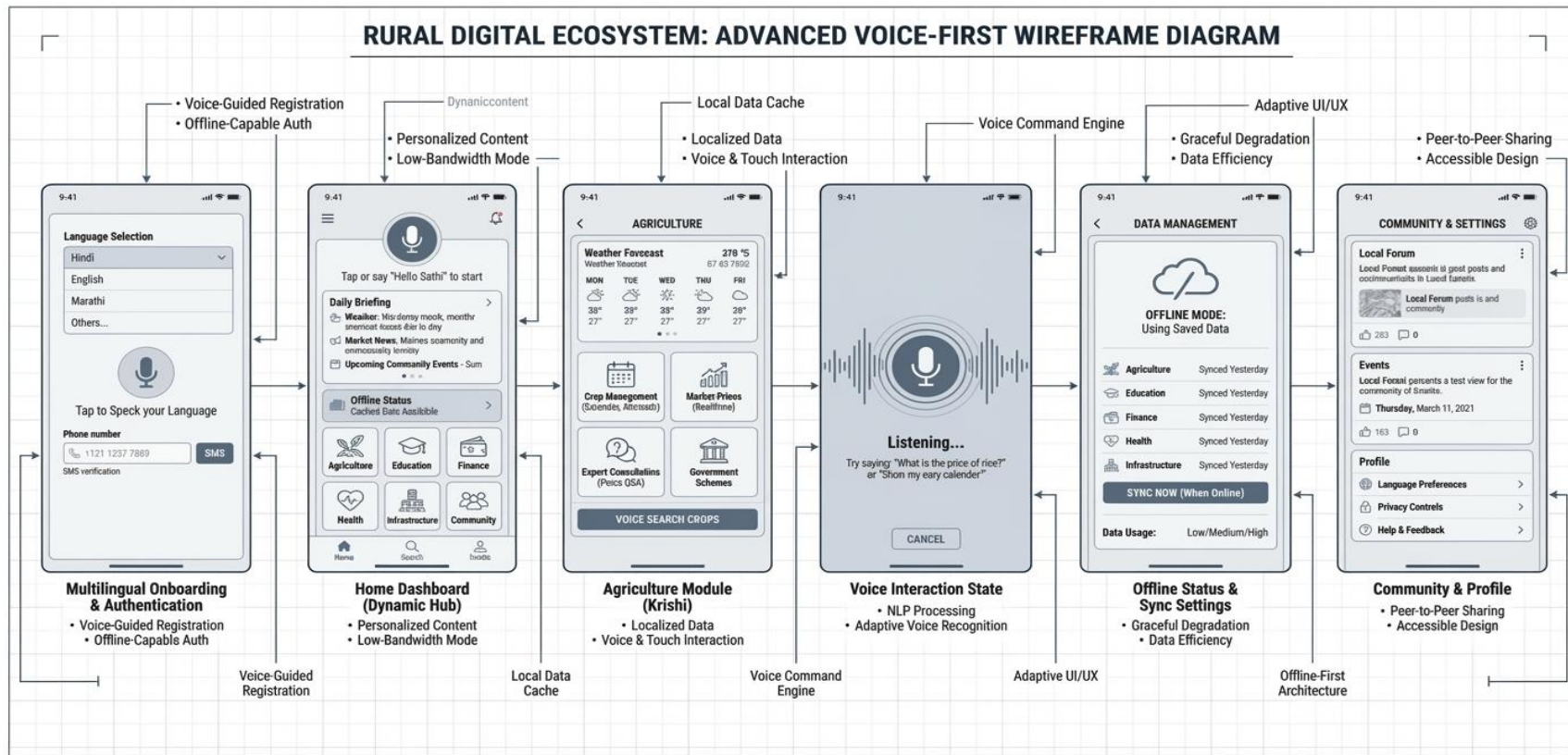
List of features offered by the solution

#	Requirement	Screens	Key Features
1	Voice-First Interface	AskScreen (hub)	12+ Indian languages, STT/TTS, persistent memory, context-aware AI. Hindi, English, Tamil, Telugu, Bengali, Marathi, Gujarati, Kannada, Malayalam, Punjabi, Odia, Assamese.
2	Language Support	LanguageSelectScreen	
3	User Authentication	LoginScreen, ProfileScreen	
4	Offline-First	All screens	Cognito phone auth, JWT, secure token storage. React Query caching, automated retry logic on reconnect.
5	Agri Supply Chain	AgriMarket, MarketPrices, Orders, BargainingGroups	Live mandi prices (25+ crops), buyer matching, collective bargaining, logistics estimation.
6	Precision Agriculture	PracticeLog, AlertsScreen	AI image analysis for pest/disease detection, carbon scoring, weather advisory.
7	Knowledge & Learning	KnowledgeDashboard, CourseDetail, PeerGroupDetail	Voice-based learning, YouTube/article integration, DigiLocker, AI recommendations.
8	Economic Services	Eligibility, SchemesList, SavingsNudge	Govt schemes (PM-KISAN, KCC, PMFBY), eligibility assessment, savings/insurance.
9	Health Services	HealthDashboard, SymptomChecker	AI screening, medical report analysis (X-ray, MRI, Path), govt health scheme links.
10	Community	VoiceRooms, VoiceRoom, CommunityScreen	Live voice rooms (Agora), peer-to-peer communities, real-time chat.
11	Open Data	SyncStatusScreen	Data export capabilities and transparent audit trails.

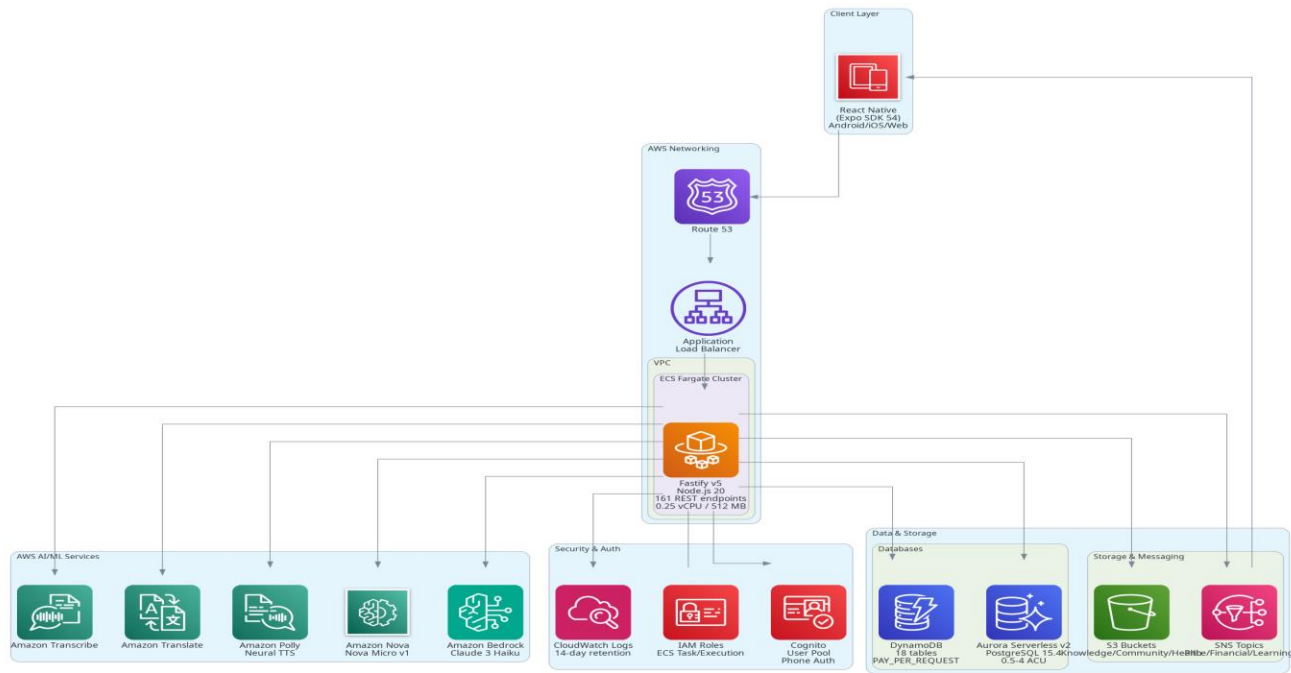
Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution

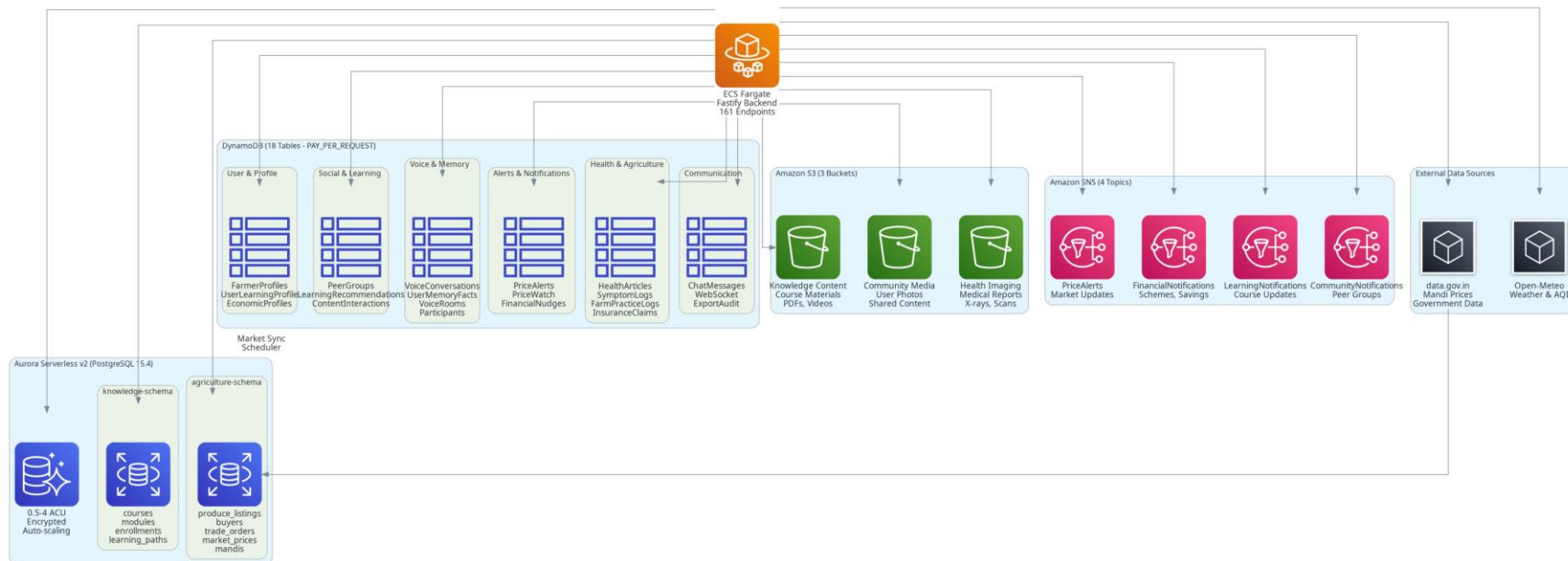


Architecture diagram of the proposed solution:



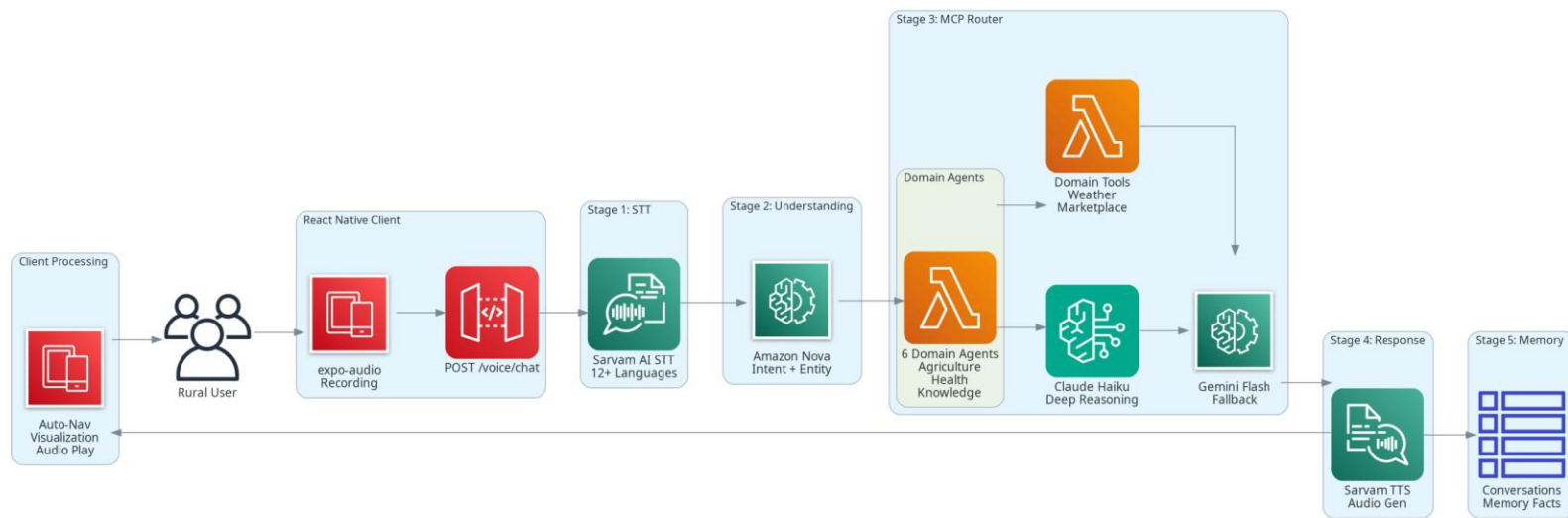
Rural Voice-First Platform - AWS Architecture

Data Architecture diagram:



Data Architecture - Storage & Databases

Voice AI pipeline diagram:



Voice-AI Pipeline Flow

Technologies utilized in the solution:

1. Cloud & Infrastructure

- Amazon Web Services (AWS) – Core cloud platform
- Amazon ECS Fargate – Containerized backend hosting
- AWS Application Load Balancer – Traffic routing & health checks
- Docker – Containerization
- AWS CloudFormation + AWS Serverless Application Model – Infrastructure as Code

2. Data Layer

- Amazon Aurora Serverless v2 (PostgreSQL) – Relational data
- Amazon DynamoDB – User state & AI memory
- Amazon S3 – Media and document storage
- Encryption at rest and in transit

3. Backend & Processing

- Node.js + Fastify – High-performance API server
- Multipart audio uploads and async processing
- Government Open Data APIs for market prices
- Amazon SNS – Alerts and notifications

4. AI & Language Services

- Amazon Bedrock (Claude 3 Haiku) – Conversational AI
- Amazon Nova – Language detection & intent routing

- Sarvam AI – Indic STT/TTS and transliteration
- Amazon Polly – Neural text-to-speech
- Amazon Translate – Multilingual translation
- Google Gemini – Fallback LLM

5. DevOps & Monitoring

- GitHub – Source control
- Docker Compose – Local development
- Amazon CloudWatch – Logging and monitoring
- Jest – Unit testing

6. Mobile Application

- React Native + Expo – Cross-platform mobile app
- TypeScript – Type-safe development
- React Navigation – Navigation system
- Expo audio/video modules – Voice recording & playback
- Secure token storage and offline support

7. Security

- Amazon Cognito – Phone-based authentication
- JWT authorization
- API rate limiting & security headers
- TLS encryption and secure storage

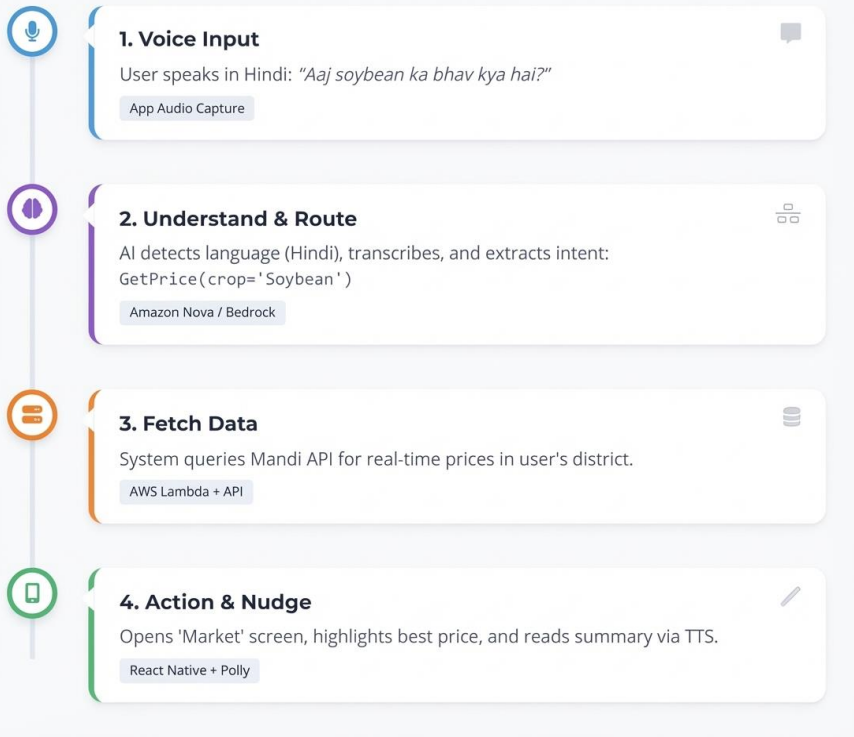
Estimated implementation cost (optional):

Requirement	Screens
Voice-First Interface	AskScreen (Hub)
Language Support	LanguageSelectScreen
User Authentication	LoginScreen, ProfileScreen
Offline-First Design	Global (All Screens)
Agri Supply Chain	AgriMarket, MarketPrices, Orders, Logistics, BargainingGroups
Precision Agriculture	PracticeLog, AlertsScreen
Knowledge & Learning	KnowledgeDashboard, CourseDetail, PeerGroupDetail
Economic Services	Eligibility, SchemesList, SavingsNudge, InsuranceClaims
Health Services	HealthDashboard, SymptomChecker
Community	VoiceRooms, CommunityScreen
Open Data	SyncStatusScreen

Key Features & Capabilities

- 12+ Indian languages, STT/TTS, persistent memory, and context-aware intent routing.
- Full localization for Hindi, English, Tamil, Telugu, Bengali, Marathi, Gujarati, Kannada, Malayalam, Punjabi, Odia, and Assamese.
- Secure AWS Cognito phone authentication, JWT, and encrypted token storage.
- React Query caching and automated retry-on-reconnect for low-bandwidth 2G/3G areas.**
- Live mandi prices (25+ crops), AI-driven buyer matching, and collective bargaining tools.
- AI image analysis for pest/disease detection, carbon scoring, and proactive weather advisories.
- Voice-based learning, YouTube/article indexing, DigiLocker verification, and AI-driven skill recommendations.
- Automated eligibility for PM-KISAN/KCC/PMFBY, AI savings "nudges," and insurance claim assistance.
- AI symptom screening, medical report analysis (X-ray, MRI, Pathology), and eSanjeevani integration.
- Live interactive voice rooms via Agora, peer communities, and multimodal chat.
- Transparent data export and full audit trails for user-controlled data sharing.

Snapshots of the prototype:



Agriculture Screen



Economics Screen



Prototype Performance report/Benchmarking:

Column1	Column2	Column3
Resource	Configuration / Usage	Estimated Monthly Cost
ECS Fargate	1 task (0.25 vCPU, 512 MB RAM)	~\$9.50
Aurora Serverless v2	0.5 ACU baseline	~\$22
DynamoDB	PAY_PER_REQUEST, moderate usage	~\$5
Application Load Balancer (ALB)	Fixed hourly cost + LCU usage	~\$18
Amazon S3	< 10 GB storage	~\$1
Amazon SNS	< 100K notifications/month	~\$1
CloudWatch Logs	14-day retention	~\$3
Amazon Cognito	< 50K MAU (Free Tier)	\$0
Bedrock Claude Haiku	Per-token inference usage	~\$10 – \$20
Nova Micro	Per-token inference usage	~\$3 – \$5
Total Estimated Cost		~\$73 – \$85 / month

Stage	~3–6 seconds
Audio upload (client → ALB → ECS)	~3–6 seconds
Sarvam STT (speech → text)	~3–6 seconds
Nova intent/language detection	~3–6 seconds
LLM reasoning (Claude/Gemini)	~3–6 seconds
Sarvam TTS (text → speech)	~3–6 seconds
Total end-to-end	~3–6 seconds

Additional Details/Future Development (if any):

Feature	Description
Real-time market updates	WebSocket-based live mandi price updates and price alerts for farmers. Alerts for crop prices, government scheme deadlines, and learning reminders. Production DigiLocker API for verified farmer identity and credentials. On-device speech recognition for areas with poor or no internet connectivity. Video-based doctor consultations for rural health support. UPI-based payment integration for agricultural trade settlements. Hyperlocal weather insights and farming advisories. Cross-district and cross-state mandi price comparison. Crop health analysis using satellite imagery. AI-assisted crop damage detection and automated claim filing. Direct listing and trading on the National Agriculture Market (e-NAM). Creditworthiness assessment for farmers based on farming activity and transactions. USSD / IVR access for rural users without smartphones. Fine-tuned Indic AI models for agriculture and rural advisory. Digital tools for Farmer Producer Organizations for collective selling and logistics.
Push notifications	
DigiLocker integration	
Offline voice support	
Telemedicine consultations	
Marketplace payments	
Weather data integration	
Multi-region price comparison	
Satellite crop monitoring	
Insurance claim automation	
e-NAM integration	
AI credit scoring	
Feature phone support	
Regional language AI models	
FPO tools	

Prototype Assets:

- GitHub Public Repository Link : <https://github.com/JAmanOG/aws-ai4bharat-hackathon>
- Demo Video Link (Max: 3 Minutes) : https://drive.google.com/file/d/1Goe-CNFJ-4YeB8auV_xDWaF8C-35eLGh/view?usp=sharing

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Thank You

